

EXECUTIVE SUMMARY

HealthConnect Final Analytics & Decision Support: Executive Summary

The HealthConnect Data Analytics track progressed from data preparation and exploratory analysis in Week 5 through deeper analysis in Week 6 and analytical validation and dashboard refinement in Week 7. Week 8 consolidates these activities into a final decision-support output based on validated analytical findings.

Analysis of **5,000 appointments** identified **2,423 no-shows (48.46%)**, **2,314 attended appointments (46.28%)**, and **263 cancellations (5.26%)**. The analysis identified several important patterns associated with appointment outcomes. Patients with previous no-show behaviour recorded a higher observed no-show rate (**55.41%**) than patients without previous no-shows (**43.51%**). No-show rates also increased across booking-lead groups, from **27.81% for appointments booked 0–7 days in advance to 60.49% for appointments booked 31–60 days in advance**.

Distance to the clinic also produced an important segmentation finding, with appointments in the **above-30 km group recording an observed no-show rate of approximately 68.06%**. Reminder-channel analysis showed differences in observed no-show rates across reminder groups, while appointment-type analysis also showed variation between follow-up, diagnostic, specialist and general consultations.

These findings provide HealthConnect with potential areas for operational investigation, particularly previous no-show behaviour, long booking lead times and longer-distance appointments. Recommendations therefore focus on developing risk-based appointment segmentation, investigating enhanced support for higher-risk segments, reviewing reminder strategies and providing validated analytical findings to the Data Science track for predictive modelling.

The analysis remains subject to important limitations. The dataset is observational and fictional/anonymized, and the observed relationships do not establish causality. Differences between reminder channels, distance groups and other segments may also reflect underlying differences between patient or appointment groups. Consequently, analytical findings should support investigation and decision-making rather than be interpreted as proof of causal effects.

A KPI validation issue identified during Week 7 was also addressed for finalisation: the **5.26% figure should represent the cancellation rate, while the correct no-show rate is 48.46%** based on 2,423 no-shows out of 5,000 appointments.

The final Data Analytics contribution is intended to support the wider HealthConnect solution by providing validated findings to Data Science for model interpretation and to Project Management for evidence-based business recommendations. The Week 8 objective is therefore not to repeat the earlier EDA, but to present a validated, interpretable and professionally documented decision-support output.